

Mathematical Methods for Finance

Module 1 – Completeness, Compactness and Normed Spaces

Lecture 4: Complete solutions to the exercises

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Contents and scope

All exercises stated in the lecture

- ▶ Block 1: contractions and Baire, Exercises A–C.
- ▶ Block 2: compactness, Exercises A–C.
- ▶ Block 3: norms and inequalities, Exercises A–C.
- ▶ Additional in-slide exercises: iteration, Baire conceptual check, extrema, three norms and sequence membership.

Exercise pack references

The numbered after-class exercises require the separate exercise pack. Their statements are not included in the supplied lecture.

Module 1

Block 1: solutions

Block 1 – Exercise A: a contraction

Exercise: A

Show that $T(x) = \cos(x)/3$ is a contraction on $\mathbb{R}$.

Definition and self-map

A contraction satisfies $|T(x) - T(y)| \leq q|x - y|$ for all x, y , with a fixed $q < 1$.

$$-\frac{1}{3} \leq T(x) \leq \frac{1}{3},$$

so T maps $\mathbb{R}$ into $\mathbb{R}$.

Solution A: the uniform estimate

Mean value theorem

If $x = y$, the desired inequality is immediate. If $x \neq y$, the mean value theorem applied to cosine gives a point c between x and y such that

$$\cos x - \cos y = -\sin(c)(x - y).$$

Thus

$$|T(x) - T(y)| = \frac{1}{3} |\sin(c)| |x - y| \leq \frac{1}{3} |x - y|.$$

Conclusion

T is a contraction with $q = 1/3 < 1$. Since $\mathbb{R}$ is complete, it has a unique fixed point.

Block 1 – Exercise B: no contradiction

Exercise: B

Explain why $T(x) = x/2$ on $(0, 1)$ does not contradict the contraction theorem.

Hypotheses to check

For $0 < x < 1$, $0 < x/2 < 1$, so $T : (0, 1) \rightarrow (0, 1)$.

$$|T(x) - T(y)| = \frac{1}{2}|x - y|.$$

The map is a contraction. We must also check completeness of the space.

Solution B: the space is incomplete

A Cauchy sequence with a missing limit

Take $u_n = 1/(n+1) \in (0, 1)$. Given $\varepsilon > 0$, choose N with $2/(N+1) < \varepsilon$. For $m, n \geq N$,

$$|u_n - u_m| \leq \frac{1}{n+1} + \frac{1}{m+1} \leq \frac{2}{N+1} < \varepsilon.$$

Thus (u_n) is Cauchy, but $u_n \rightarrow 0$ in $\mathbb{R}$, and $0 \notin (0, 1)$. A limit inside $(0, 1)$ would also be its limit in $\mathbb{R}$, contradicting uniqueness of limits.

Solution B: fixed point and iteration

The fixed point is outside the domain

The fixed point equation gives

$$x = \frac{x}{2} \iff x = 0.$$

There is no fixed point in $(0, 1)$. Starting from $x_0 \in (0, 1)$,

$$x_n = \frac{x_0}{2^n} \longrightarrow 0.$$

Conclusion

The completeness hypothesis fails. Banach's theorem therefore does not apply, so there is no contradiction.

Block 1 – Exercise C: nowhere dense

Exercise: C

Give an example of a nowhere dense subset of $\mathbb{R}$.

Definition

A subset $A \subset \mathbb{R}$ is nowhere dense when

$$\text{int}(\overline{A}) = \emptyset.$$

We must take the closure before checking its interior.

Solution C: a singleton

Choose $A = \{0\}$

The set is closed because its complement $(-\infty, 0) \cup (0, \infty)$ is open. Hence $\overline{A} = A$. For any $r > 0$, the interval $(-r, r)$ contains $r/2 \neq 0$, so it is not contained in A . Therefore 0 is not an interior point and

$$\text{int}(\overline{A}) = \text{int}(\{0\}) = \emptyset.$$

Conclusion

$\{0\}$ is nowhere dense in $\mathbb{R}$. By contrast, $\mathbb{Q}$ has empty interior but is not nowhere dense because $\overline{\mathbb{Q}} = \mathbb{R}$.

Module 1

Block 2: solutions

Block 2 – Exercise A: the interval $[2, 5]$

Exercise: A

Decide whether $[2, 5]$ is compact in $\mathbb{R}$.

Heine–Borel

A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. Here the ambient space is $\mathbb{R}$.

Solution A: closed and bounded

Closedness

The complement is

$$\mathbb{R} \setminus [2, 5] = (-\infty, 2) \cup (5, \infty),$$

which is open. Thus $[2, 5]$ is closed.

Boundedness and conclusion

For every $x \in [2, 5]$, $|x| \leq 5$. Hence the interval is bounded. By Heine–Borel,

$[2, 5]$ is compact in $\mathbb{R}$.

Block 2 – Exercise B: the open unit disk

Exercise: B

Decide whether

$$D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$$

is compact.

Boundedness

Every $x \in D$ has $\|x\|_2 < 1$, so D is bounded. We must also check closedness.

Solution B: a missing boundary point

A sequence in the disk

Take

$$u_n = \left(1 - \frac{1}{n}, 0\right), \quad n \geq 1.$$

Since $0 \leq 1 - 1/n < 1$, we have $u_n \in D$. Moreover,

$$\|u_n - (1, 0)\|_2 = \frac{1}{n} \longrightarrow 0.$$

The limit $(1, 0)$ does not belong to D , because $1^2 + 0^2 = 1$.

Conclusion

D is not closed and therefore is not compact by Heine–Borel.

Solution B: the subsequence viewpoint

Failure of sequential compactness

Every subsequence (u_{n_k}) also satisfies

$$\|u_{n_k} - (1, 0)\|_2 = \frac{1}{n_k} \longrightarrow 0.$$

Thus every subsequence has the same limit outside D . No subsequence can converge to a point of D , because limits in $\mathbb{R}^2$ are unique.

What this shows

Boundedness alone does not imply compactness. The closed unit disk, with ≤ 1 instead of < 1 , is compact.

Block 2 – Exercise C: unit vectors in ℓ^2

Exercise: C

Explain why the sequence (e_n) in ℓ^2 has no convergent subsequence.

The vectors and their norms

The vector e_n has a 1 in coordinate n and zeros elsewhere.

$$\|e_n\|_2 = \left(\sum_{j=1}^{\infty} |(e_n)_j|^2 \right)^{1/2} = 1.$$

The sequence is bounded.

Solution C: distinct vectors stay apart

Distance computation

For $n \neq m$, $e_n - e_m$ has one coordinate equal to 1, one equal to -1 , and all others zero. Hence

$$\|e_n - e_m\|_2^2 = 1^2 + (-1)^2 = 2, \quad \|e_n - e_m\|_2 = \sqrt{2}.$$

No Cauchy subsequence

For any subsequence (e_{n_k}) , all distinct indices give distance $\sqrt{2}$. Taking $\varepsilon = 1$, for every K choose $k = K$, $\ell = K + 1$. Then

$$\|e_{n_k} - e_{n_\ell}\|_2 = \sqrt{2} > 1.$$

Solution C: convergence and compactness

Conclusion

Every convergent sequence in a metric space is Cauchy. Since no subsequence of (e_n) is Cauchy, no subsequence is convergent.

Closed and bounded need not be compact

Every e_n belongs to the closed unit ball

$$B = \{x \in \ell^2 : \|x\|_2 \leq 1\}.$$

The sequence has no convergent subsequence, so B is not compact. Heine–Borel's closed-and-bounded criterion applies to $\mathbb{R}^n$, not to arbitrary infinite-dimensional normed spaces.

Module 1

Block 3: solutions

Block 3 – Exercise A: three norms

Exercise: A

Compute $\|(1, -2, 2)\|_1$, $\|(1, -2, 2)\|_2$ and $\|(1, -2, 2)\|_\infty$.

Definitions

For $x = (x_1, x_2, x_3)$,

$$\|x\|_1 = |x_1| + |x_2| + |x_3|,$$

$$\|x\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + |x_3|^2}, \quad \|x\|_\infty = \max_j |x_j|.$$

Solution A: computations

Substitution

For $x = (1, -2, 2)$,

$$\|x\|_1 = |1| + |-2| + |2| = 1 + 2 + 2 = 5,$$

$$\|x\|_2 = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = 3,$$

$$\|x\|_\infty = \max\{1, 2, 2\} = 2.$$

Result

$$\|x\|_1 = 5, \quad \|x\|_2 = 3, \quad \|x\|_\infty = 2.$$

Block 3 – Exercise B: Cauchy–Schwarz

Exercise: B

Use Cauchy–Schwarz to bound $|x_1 + 2x_2 + 3x_3|$ in terms of $\|x\|_2$.

Choose the second vector

Let $a = (1, 2, 3)$. Then

$$x_1 + 2x_2 + 3x_3 = \langle x, a \rangle.$$

Cauchy–Schwarz gives $|\langle x, a \rangle| \leq \|x\|_2 \|a\|_2$.

$\langle x, y \rangle$ is a product scalar.

$$\langle x, y \rangle = x \cdot y = \sum_{i=1}^3 x_i y_i.$$

Solution B: the bound and its sharpness

Compute the constant

Since

$$\|a\|_2 = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14},$$

we obtain

$$|x_1 + 2x_2 + 3x_3| \leq \sqrt{14} \|x\|_2.$$

Equality

Equality holds when x is a scalar multiple of $(1, 2, 3)$, including $x = 0$. Taking $x = a$ yields $14 = \sqrt{14}\sqrt{14}$, so the constant cannot be reduced.

Block 3 – Exercise C: comparison of norms

Exercise: C

Show that $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ in $\mathbb{R}^n$.

First inequality

For each j ,

$$|x_j|^2 \leq \sum_{i=1}^n |x_i|^2 = \|x\|_2^2.$$

Taking non-negative square roots and then the maximum gives

$$\boxed{\|x\|_\infty \leq \|x\|_2.}$$

Solution C: the second inequality

Expand the square

All absolute values are non-negative, so

$$\left(\sum_{i=1}^n |x_i| \right)^2 = \sum_{i=1}^n |x_i|^2 + 2 \sum_{i < j} |x_i| |x_j| \geq \sum_{i=1}^n |x_i|^2.$$

Taking non-negative square roots gives

$$\|x\|_2 \leq \|x\|_1.$$

Conclusion

Together, $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ for every $x \in \mathbb{R}^n$.

Module 1

Additional exercises in the lecture

Iteration: the first three steps

Exercise: Iteration – lecture slide 9

For $T(x) = x/2 + 1$, starting from $x_0 = 0$, compute x_1, x_2, x_3 and observe convergence to 2.

Direct calculation

$$x_1 = T(0) = 1,$$

$$x_2 = T(1) = \frac{1}{2} + 1 = \frac{3}{2},$$

$$x_3 = T\left(\frac{3}{2}\right) = \frac{3}{4} + 1 = \frac{7}{4}.$$

Iteration: formula and convergence

Error recurrence

Since the fixed point is 2,

$$x_{n+1} - 2 = \frac{1}{2}(x_n - 2).$$

As $x_0 - 2 = -2$, induction gives

$$x_n - 2 = -2 \left(\frac{1}{2}\right)^n, \quad x_n = 2 - 2^{1-n}.$$

Limit

The error satisfies $|x_n - 2| = 2^{1-n} \rightarrow 0$. Thus $x_n \rightarrow 2$, and each iteration halves the error.

Baire: why completeness is natural

Exercise: Conceptual check – lecture slide 13

Why is completeness the natural hypothesis in Baire's theorem?

The proof mechanism

Starting inside a non-empty open set, one constructs nested closed balls avoiding successive closed nowhere dense sets. Their radii tend to zero. The centers form a Cauchy sequence. Completeness supplies a limit in the space, and closedness places it inside every ball. Thus the limit avoids every set in the countable family.

Baire: an incomplete counterexample

The space $\mathbb{Q}$ with its usual metric

Enumerate its points as $\mathbb{Q} = \{q_1, q_2, \dots\}$. Then

$$\mathbb{Q} = \bigcup_{n=1}^{\infty} \{q_n\}.$$

Each singleton is closed in $\mathbb{Q}$. Its relative interior is empty because every ball centered at a rational contains another rational. Thus each singleton is nowhere dense *in* $\mathbb{Q}$.

Conclusion

The incomplete space $\mathbb{Q}$ is a countable union of closed nowhere dense subsets of itself. Completeness guarantees Baire's conclusion, though some incomplete spaces also have the Baire property.

Extreme values: $f(x) = 1/x$ on $(0, 1)$

Exercise: Example – lecture slide 20

Does $f(x) = 1/x$ attain a maximum on $(0, 1)$? Which hypothesis fails?

No maximum

For any $x \in (0, 1)$, the point $x/2$ also belongs to $(0, 1)$ and

$$f(x/2) = \frac{2}{x} > \frac{1}{x} = f(x).$$

Hence no point can maximize f . Also, for $x_n = 1/(n+1)$,

$$f(x_n) = n+1 \longrightarrow +\infty.$$

Extreme values: the missing hypothesis

Continuity holds

$f(x) = 1/x$ is continuous at every point of $(0,1)$ because the denominator is non-zero there.

Compactness fails

The interval $(0,1)$ is bounded but not closed in $\mathbb{R}$, so it is not compact. The extreme value theorem requires both continuity and compactness of the domain. Therefore it does not apply.

Conclusion

f has no maximum on $(0,1)$. Its infimum is 1, but it also has no minimum because $x = 1$ is excluded.

Additional norm computation

Exercise: Compute – lecture slide 28

For $x = (2, -1, 3)$, compute $\|x\|_1$, $\|x\|_2$ and $\|x\|_\infty$.

All steps

$$\|x\|_1 = |2| + |-1| + |3| = 2 + 1 + 3 = 6,$$

$$\|x\|_2 = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14},$$

$$\|x\|_\infty = \max\{2, 1, 3\} = 3.$$

The comparison from Block 3 gives $3 \leq \sqrt{14} \leq 6$.

Sequence membership: $x_i = 1/i$

Exercise: Membership – lecture slide 32

For $x = (1, 1/2, 1/3, \dots)$, decide whether $x \in \ell^1$, ℓ^2 and ℓ^∞ .

Definitions

$$x \in \ell^1 \iff \sum_{i=1}^{\infty} |x_i| < \infty,$$

$$x \in \ell^2 \iff \sum_{i=1}^{\infty} |x_i|^2 < \infty, \quad x \in \ell^\infty \iff \sup_i |x_i| < \infty.$$

Membership: divergence in ℓ^1

Group the harmonic series

For each $k \geq 1$, the indices $2^{k-1} + 1, \dots, 2^k$ contain 2^{k-1} terms. Each term is at least $1/2^k$, so

$$\sum_{i=2^{k-1}+1}^{2^k} \frac{1}{i} \geq \frac{2^{k-1}}{2^k} = \frac{1}{2}.$$

Consequently,

$$\sum_{i=1}^{2^K} \frac{1}{i} \geq 1 + \frac{K}{2} \rightarrow +\infty.$$

Therefore $\sum_i |x_i| = \infty$, hence $x \notin \ell^1$.

Membership: convergence in ℓ^2

A telescoping comparison

For $i \geq 2$,

$$\frac{1}{i^2} \leq \frac{1}{i(i-1)} = \frac{1}{i-1} - \frac{1}{i}.$$

Therefore for $N \geq 2$,

$$\sum_{i=1}^N \frac{1}{i^2} \leq 1 + \sum_{i=2}^N \left(\frac{1}{i-1} - \frac{1}{i} \right) = 2 - \frac{1}{N} < 2.$$

The partial sums are increasing and bounded, so the series converges.

Conclusion

$\sum_i |x_i|^2 < \infty$, hence $x \in \ell^2$.

Membership: boundedness in ℓ^∞

Supremum

For every $i \geq 1$, $0 < 1/i \leq 1$, and equality holds at $i = 1$. Hence

$$\sup_{i \geq 1} \frac{1}{i} = 1 < \infty.$$

Thus $x \in \ell^\infty$ and $\|x\|_\infty = 1$.

Final classification

$$x \notin \ell^1, \quad x \in \ell^2, \quad x \in \ell^\infty.$$

Membership depends on the summability condition, not merely on whether the coordinates tend to zero.